

PAPER_513: NGC 1277 Wolfram Hypergraph Spacetime Dimension

Star Magic UQFF Framework — Session 138

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Module: source179.cpp | **Target:** NGC 1277 (compact galaxy, Perseus cluster)

Abstract

NGC 1277 hosts one of the most overmassive black holes known relative to its host galaxy: $M_{\text{BH}} \approx 17$ billion $M_{\odot}$, constituting $\approx 14\%$ of the galaxy's bulge mass. The Wolfram hypergraph BFS dimension estimator in SOURCE179 computes D_{eff} from causal graph growth rates and applies UQFF correction $\delta D = k_{\text{PCR}} \times (D_{\text{eff}} - 3)$. Near ultramassive BHs, spacetime curvature should produce measurable D_{eff} deviations above 3.

1. Hypergraph BFS Dimension Formula

$$D_{\text{eff}}(d) = \frac{\log N_{\text{BFS}}(d)}{\log d}, \quad N_{\text{BFS}}(d) \approx \min(b^d, N_{\text{total}})$$

where b = mean branching factor (4.5 for compact/curved region), d = BFS depth.

UQFF correction:

$$\delta D = k_{\text{PCR}} \cdot (D_{\text{eff}} - 3)$$

$$D_{\text{corrected}} = D_{\text{eff}} + \delta D$$

2. NGC 1277 Parameters

Parameter	Value
BH mass	$M_{\text{BH}} = 1.7\text{e}10 M_{\odot}$
Schwarzschild radius	$r_s = 50 \text{ AU}$
Host galaxy stellar mass	$M_* \approx 1.2\text{e}11 M_{\odot}$
BH/bulge ratio	$\approx 14\%$ (vs typical 0.1%)
Distance	73 Mpc (Perseus cluster)
Velocity dispersion	$\sigma \approx 333 \text{ km/s}$

3. Computed Result

For branching factor $b = 4.5$, depth $d = 6$:

$$N_{\text{BFS}}(6) = 4.5^6 \approx 8303 \Rightarrow D_{\text{eff}} = \frac{\log 8303}{\log 6} \approx 4.83$$

$$\delta D = 0.314 \times (4.83 - 3) = 0.575 \Rightarrow D_{\text{corrected}} \approx 5.40$$

A $D_{\text{corrected}} \approx 5.4$ indicates the UQFF framework interprets the extreme curvature near NGC 1277's BH as requiring 2 additional effective dimensions — consistent with 26D layer gravity framework predictions.

4. Validation

- C++ term: `SOURCE179::NGC1277_HypergraphD_Term → NGC1277_HypergraphDimension`
- CP2 class: `NGC1277HypergraphDimCalculator → D_eff, D_corrected, δD, k_PCR`

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524\text{e-}29 \text{ m}^2$	$\sigma_T = 6.6524\text{e-}29 \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Massive object (Eta Carinae/TON618/NGC1277) X-ray + optical	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \text{ luminosity} \approx g_{\text{total}} \times M_{\text{env}}$	$L_X M_{\text{BH}} \sim 10^9 - 10^{10} M_{\odot}$	Chandra/HST	Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra/HST	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Massive object (Eta Carinae/TON618/NGC1277) through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra/HST monitoring observations.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

References

- van den Bosch et al. (2012) *An overmassive black hole in the compact lenticular galaxy NGC 1277*, Nature 491, 729
- Walsh et al. (2016) *Revisiting the NGC 1277 black hole mass*, ApJ 817, 2
- Murphy, D.T. *PAPER_507: WolframFieldUnityEngine Hypergraph*